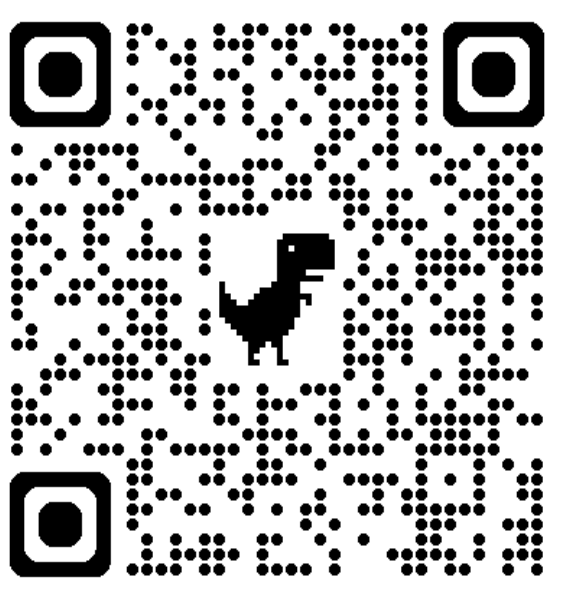


Exponentially tilted empirical likelihood sensitivity for Bayesian moment conditions models

Shelly Xie, Catherine Forbes and David Frazier



1. Introduction

- Moment condition models avoid the need to specify the full data distribution process.
- Exponentially tilted empirical likelihood (ETEL) can be interpreted probabilistically as a nonparametric likelihood.

- ETEL provides a useful framework for Bayesian moment condition models (Chib et al., 2018).
- **How does BETEL/ETEL perform in the presence of data contamination (outliers)?**

2. Methodology

Moment condition model:

$$E_P[\mathbf{g}(\mathbf{y}, \boldsymbol{\theta})] = \mathbf{0}. \quad (1)$$

- Parameter of interest $\boldsymbol{\theta} := (\theta_1, \dots, \theta_p)$.
- Observed data $\mathbf{y} = (y_1, \dots, y_n)$ from the unknown distribution P .

ETEL (Schennach, 2005):

- Defined as

$$p_{ETEL}(\mathbf{y}|\boldsymbol{\theta}) = \prod_{i=1}^n p_i(\boldsymbol{\theta}). \quad (2)$$

- By solving the constrained maximization problem:

$$\max_{p_1, \dots, p_n} \sum_{i=1}^n [-p_i \log(np_i)] \quad (3)$$

subject to

$$\sum_{i=1}^n p_i(\boldsymbol{\theta}) = 1 \text{ and } \sum_{i=1}^n p_i(\boldsymbol{\theta}) \mathbf{g}(y_i; \boldsymbol{\theta}) = \mathbf{0}. \quad (4)$$

- The ETEL probabilities:

$$p_i(\boldsymbol{\theta}) := \frac{\exp(\hat{\boldsymbol{\lambda}}(\boldsymbol{\theta})^T \mathbf{g}(y_i; \boldsymbol{\theta}))}{\sum_{j=1}^n \exp(\hat{\boldsymbol{\lambda}}(\boldsymbol{\theta})^T \mathbf{g}(y_j; \boldsymbol{\theta}))} \quad (5)$$

where the Lagrange multipliers $\hat{\boldsymbol{\lambda}}(\boldsymbol{\theta})$ is given by

$$\hat{\boldsymbol{\lambda}}(\boldsymbol{\theta}) = \argmin_{\boldsymbol{\lambda}} \sum_{i=1}^n \exp(\boldsymbol{\lambda}^T \mathbf{g}(y_i; \boldsymbol{\theta})). \quad (6)$$

Bayesian ETEL (BETEL) (Schennach, 2005):

- Through Bayes' theorem, the BETEL posterior takes the form:

$$\pi_{BETEL} \propto p_{ETEL}(\mathbf{y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}). \quad (7)$$

3. Set up

Example 1 Simple linear regression:

$$y_i = \beta_0 + \beta_1 x_i + e_i \text{ for } i = 1, \dots, 1000$$

- Three moment conditions:

$$E_P[e_i] = 0, \quad E_P[x_i e_i] = 0, \quad E_P[e_i^3] = 0. \quad (8)$$

- True parameter values: $\beta_0 = 2, \beta_1 = -3$.

- Data generating process:

- $x_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(1, 1)$.
- To introduce outliers, consider the contaminated model:

$$H_e = (1 - c)F + cG \quad (9)$$

where $F = \mathcal{N}(0, 1)$ and $G = \mathcal{N}(5, 1)$. Fix contamination rate at $c = 0.10$.

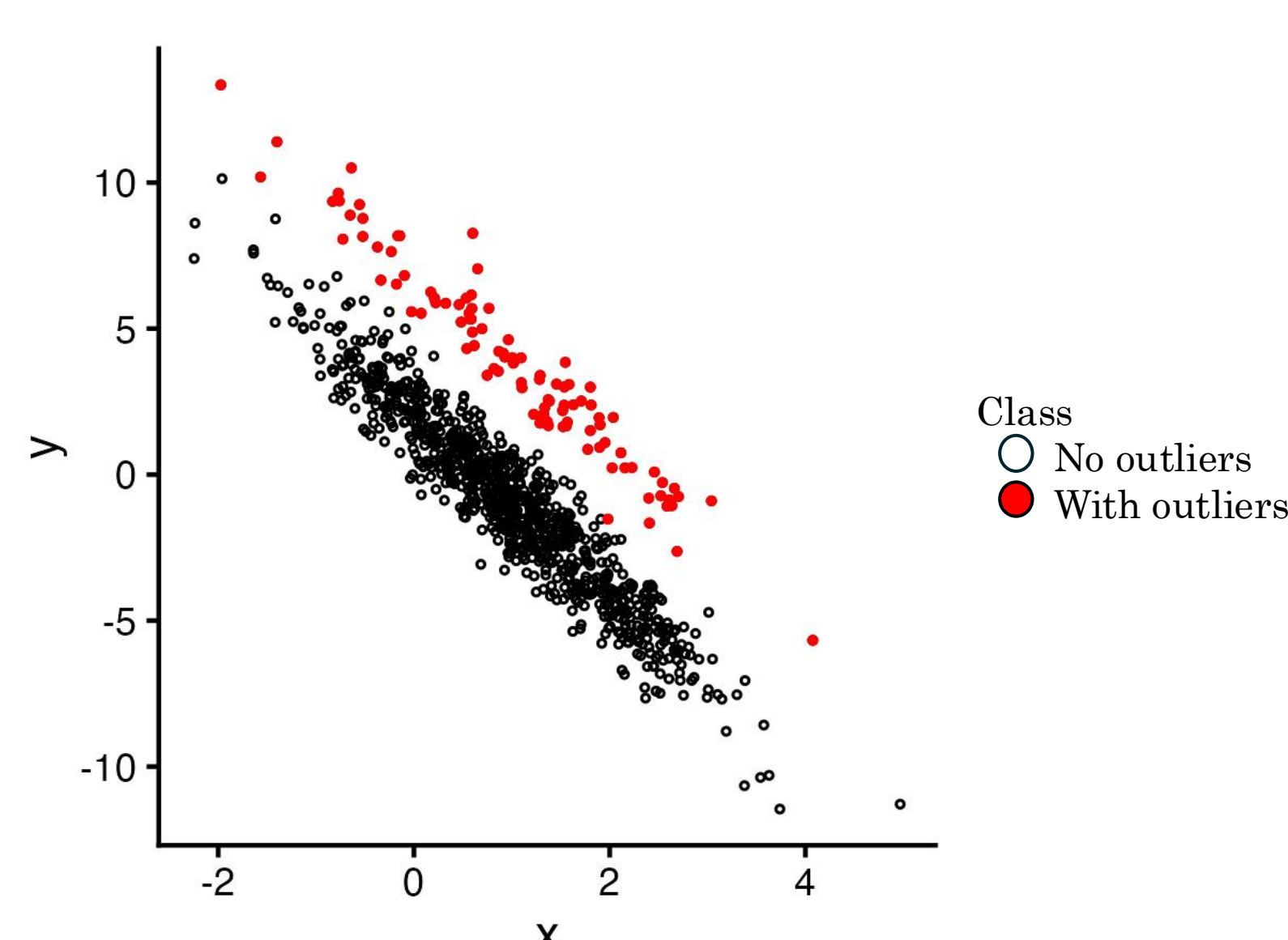


Figure 1: Simulated data values. Black points are normal observations with error from distribution F and red points represent outliers from contaminated distribution G .

4. Research gap

MCMC:

- Metropolis-Hasting within Gibbs.
- Sample $\boldsymbol{\theta}$ jointly.
- 50 replications; 50,000 draws; 20,000 burn-in.

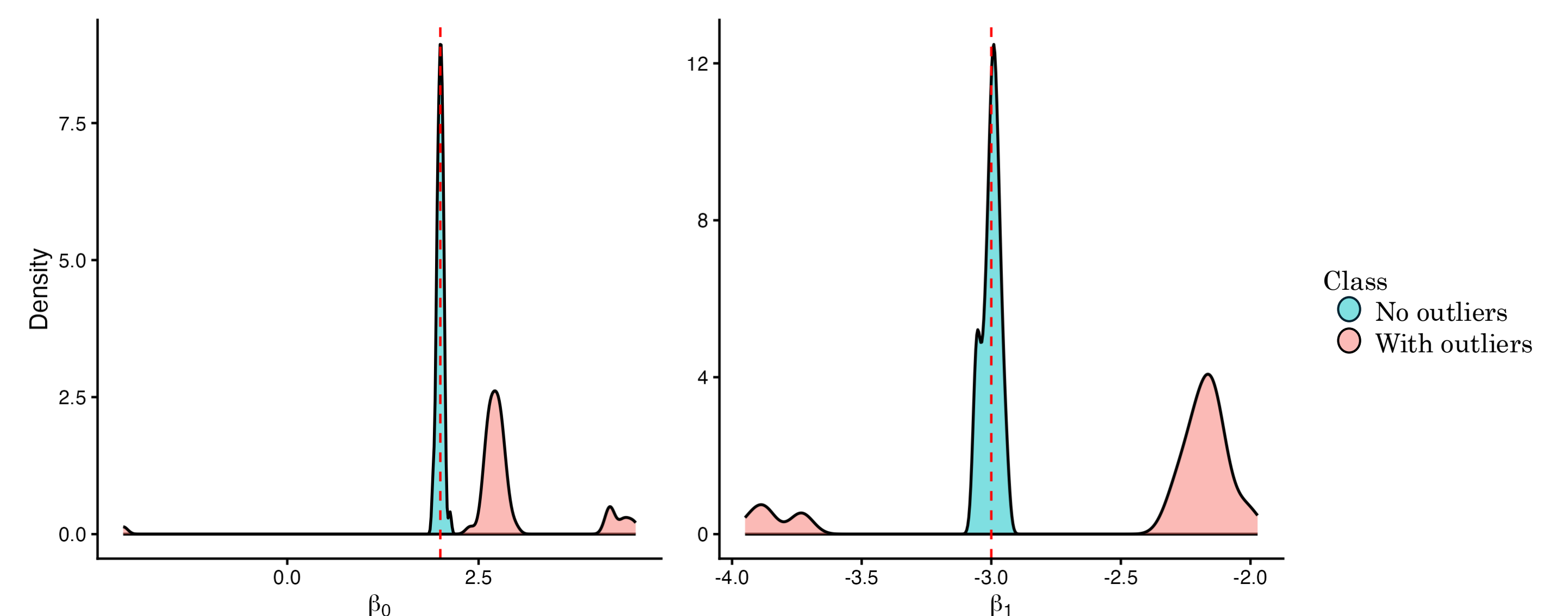


Figure 2: Posterior mean density plot. The vertical dashed line indicates the true parameter values. The blue distribution corresponds to uncontaminated data, while red corresponds to data with 10% contamination.

Define $\mathbf{Z}$ as weighted residuals:

$$Z_i(\boldsymbol{\theta}_{ETEL}) = p_i(\boldsymbol{\theta}_{ETEL}) \mathbf{g}_i(\boldsymbol{\theta}_{ETEL}), \quad (10)$$

$$Z_i(\boldsymbol{\theta}_{GMM}) = 1/n \mathbf{g}_i(\boldsymbol{\theta}_{GMM}). \quad (11)$$

Under regularity condition:

$$Z_i(\hat{\boldsymbol{\theta}}_{ETEL}) = \frac{1}{n} \mathbf{g}_i(\boldsymbol{\theta}_0) + \frac{1}{n} \frac{\partial \mathbf{g}_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} (\hat{\boldsymbol{\theta}}_{ETEL} - \boldsymbol{\theta}_0) + o_p(n^{-1}), \quad (12)$$

$$Z_i(\hat{\boldsymbol{\theta}}_{GMM}) = \frac{1}{n} \mathbf{g}_i(\boldsymbol{\theta}_0) + \frac{1}{n} \frac{\partial \mathbf{g}_i(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}'} \Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}_0} (\hat{\boldsymbol{\theta}}_{GMM} - \boldsymbol{\theta}_0) + o_p(n^{-1}). \quad (13)$$

Likelihood ratio test:

$$H_0: f_{ETEL}(\mathbf{Z}; \boldsymbol{\gamma}_{ETEL}) = f_{GMM}(\mathbf{Z}; \boldsymbol{\gamma}_{GMM}) \quad (14)$$

$$\Leftrightarrow H_0: \boldsymbol{\gamma}_{ETEL} = \boldsymbol{\gamma}_{GMM}$$

- Skew-normal distribution f , characterized by parameters $\boldsymbol{\gamma}$.

Scenario	N	Reps	Weighted Residuals (ETEL)				Weighted Residuals (GMM)				All Reject
			1	2	3	All	1	2	3	All	
1	100	50	0.08	0.02	0.44	0.00	0.00	0.00	0.52	0.00	0.00
2	100	50	0.96	0.96	1.00	0.92	0.96	0.98	1.00	0.94	0.92
3	500	50	0.14	0.10	0.76	0.08	0.12	0.10	0.76	0.02	0.02
4	500	50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
5	1000	50	0.16	0.02	0.60	0.02	0.14	0.04	0.60	0.00	0.00
6	1000	50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

Table 1: Rejection rates for likelihood ratio test at significance level $\alpha = 10\%$.

5. Contributions and future research

Contributions:

- Constructed a formal likelihood ratio test in the ETEL framework.
- Demonstrated that ETEL is sensitive to data contamination.

Future research:

- We aim to develop diagnostic measure to identify outliers, enabling more reliable application of ETEL in practice.

References:

- Chib, S., Shin, M., & Simoni, A. (2018). Bayesian estimation and comparison of moment condition models. *Journal of the American Statistical Association*, 113(524), 1656-1668.
- Schennach, S. M. (2005). Bayesian exponentially tilted empirical likelihood. *Biometrika*, 92(1), 31-46.